

Jose Arturo Santoyo Guilarte

Intermediate Systems Developer / Integrator

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Computer Systems Engineer with 7+ years of experience across full-stack development, DevOps, and QA automation. Currently modernizing legacy .NET line-of-business applications to ASP.NET Core 8 on Azure Kubernetes Service for the City of Ottawa. Comfortable across the stack — from C# / Oracle and CI/CD pipelines to Python audio DSP — with a strong bias toward automation, clean architecture, and reliable delivery.

SKILLS

Languages	C#, JavaScript, Python, T-SQL, PL/SQL, PowerShell, Bash, HTML, CSS
Frameworks	ASP.NET Core 8, ASP.NET MVC, Razor Pages, ASP.NET WebForms, Bootstrap
Cloud & DevOps	Azure (AKS, Azure AD / OIDC), Azure DevOps Pipelines, Docker, Kubernetes, CI/CD, Git
Data	Oracle, SQL Server, Redis
QA / Testing	Selenium WebDriver, Protractor, BDD
Audio / AI	NAudio, librosa, Demucs, WaveSurfer.js
Spoken languages	Spanish (native), English (fluent), French (working)

EXPERIENCE

Intermediate Systems Developer / Integrator

Apr 2021 – Present

City of Ottawa · Contract, Full-time · Ottawa, ON

- Enhance and automate business tasks by analyzing, designing, and implementing scripts, small programs, and database components.
- Identify and prepare functional specifications in collaboration with clients, vendors, and other ITS staff.
- Manage source code with standard SCM tools; write technical specifications and test scripts.
- Perform preventive maintenance by developing in-house upgrades and applying vendor-supplied patches.
- Modify applications and components to ensure continued operations across environments.

FEATURED PROJECT

Legacy .NET Modernization — Azure Kubernetes

Led the modernization of a 15+ year old ASP.NET WebForms line-of-business application to a cloud-ready ASP.NET Core 8 solution deployed to Azure Kubernetes Service.

- Re-implemented the legacy app on .NET 8 with a clean layered architecture (Web / Data projects, repository pattern, strongly-typed view models).
- Replaced commercial WebForms UI components with open-source equivalents (FullCalendar.js, Bootstrap), eliminating per-developer licensing costs.
- Integrated Azure AD / OpenID Connect with a database-driven role allowlist and per-session active-role enforcement.
- Containerized the app (multi-stage Dockerfile, non-root runtime) and authored Kubernetes manifests (base + Dev/QA overlays) with health probes, resource limits, and TLS ingress.
- Built an Azure DevOps YAML pipeline for automated build and multi-environment delivery (Dev → QA).

C# / .NET 8 ASP.NET Core Oracle Azure AD / OIDC Docker Kubernetes (AKS) Azure DevOps FullCalendar Bootstrap

DevOps Specialist

Jun 2018 – May 2020

Inteloom Incorporated

- Maintained and improved automation tooling, and advised on QA best practices.
- Built automated deployments using configuration management and CI/CD practices.
- Prepared modules for production with release procedures and knowledge-transfer documentation.
- Deployed new modules, upgrades, and fixes to production; handled routine application maintenance and troubleshooting.

Quality Assurance Engineer

Jun 2017 – Jun 2018

Inteloom Incorporated

- Participated in the QA process across Inteloom projects; built and maintained automated test suites using Selenium WebDriver and Protractor, applying behavior-driven development concepts.

Web Developer & Business Analyst

Feb 2017 – Oct 2017

The Handmade Bride

- Designed and developed the company's e-commerce site; handled regular updates and maintenance of the business website and online store.

SELECTED PERSONAL PROJECTS

MixReady — ASP.NET Core 8 web app that turns any song into a DJ-ready track.

Hybrid C# (NAudio) + Python (librosa, Demucs) pipeline detects BPM, key, and song structure, separates stems, and synthesizes beat-aligned intros that crossfade smoothly into the original mix.

- Interactive WaveSurfer.js waveform UI — click to pick extraction and song-start points, snapped server-side to the beat grid.
- AI stem separation with per-stem intro selection; auto-detection of BPM, key, and genre on upload.
- Resilient pipeline — every Python-powered feature has a pure-C# fallback for graceful degradation.
- Redis-backed track metadata and async job state; containerized with Docker / docker-compose.

C# / .NET 8 ASP.NET Core Razor Pages NAudio Python librosa Demucs Redis Docker WaveSurfer.js

Medical Glossary App

Cross-platform mobile glossary for medical terminology.

Liquid Flow Meter

Embedded project: sensor + microcontroller with real-time data output.

Elevator System (Rational Rose)

Real-time elevator system modelled in IBM Rational Rose RealTime using UML state machines.

EDUCATION

B.Eng. in Computer Systems Engineering

2011 – 2017

Carleton University · Department of Systems and Computer Engineering · Ottawa, ON

VOLUNTEERING & TEACHING

- VP External Affairs** — **H.O.L.A.S., Carleton University** (Sep 2014 – Apr 2015): coordinated communication with partner organizations, organized fundraising events for humanitarian programs in Latin America, and managed the H.O.L.A.S. website and social media.
- Spanish Teaching Assistant** — **Carleton University** (Sep 2013 – Mar 2014): taught Spanish in individual and group format, facilitated oral/writing practice, and marked assignments.
- Sub-Coordinator** — **Inspira Youth Leadership Conference** (2016): managed executive team, logistics, and resources for the volunteer staff.
- Instructor** — **Virtual Ventures** (2013): assisted younger students in computer and robotics workshops.